

limbs of  $a, b$   
(base  $10^{18}$ )



pad to length  $n$   
 $n \geq \ell a + \ell b - 1$



forward NTT  
 $\text{mod } p_0, p_1, p_2$



pointwise  
 $\hat{c} = \hat{a} \cdot \hat{b} \text{ mod } p_i$



inverse NTT  
 $\times n^{-1} \text{ mod } p_i$



Garner:  
 $3 \text{ residues} \rightarrow c_k < p_0 p_1 p_2$



carry:  
 $c_k \rightarrow \text{base-}10^{18} \text{ limbs}$